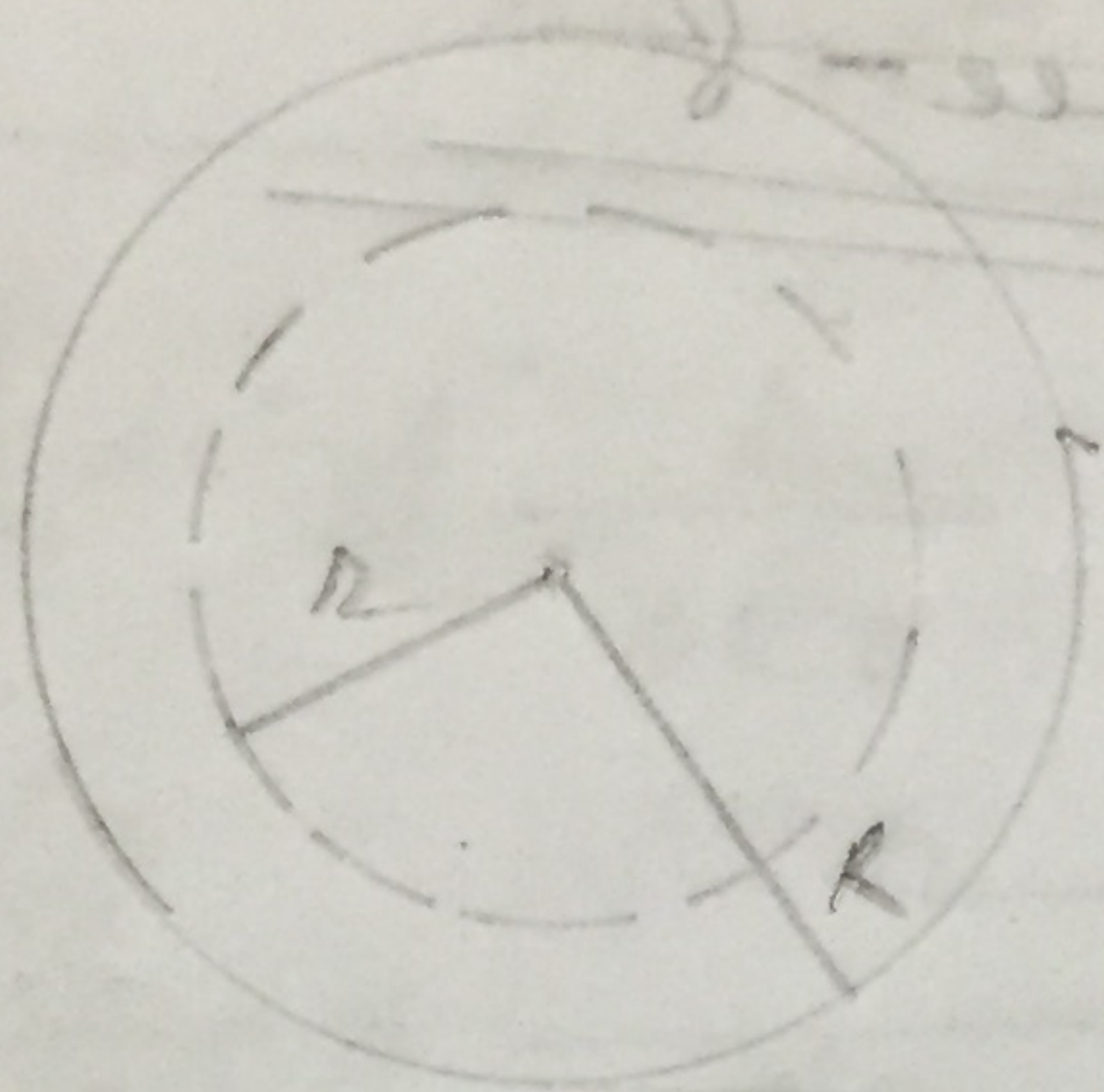


Ans. to the Q. 01

(a)



To find the electric field at ~~points~~ ~~both~~ inside of the thick conducting rod, we can use Gauss's Law.

Inside the rod, $r < R$ and length L , the total charge enclosed by the cylinder is the charge density ρ times the surface area of the cylinder,

$$Q_{enc} = \rho \cdot A \quad [A = \text{surface area of cylinder}]$$

The electric field $\vec{E}$ is radial and has the same magnitude at all points on the cylindrical surface due to the symmetry of the problem.

therefore,

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow \oint E \cdot dA \cos \phi_1 + \oint E \cdot dA \cos \phi_2 + \oint E \cdot dA \cos \phi_3 = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow \oint E \cdot dA \cos 90^\circ + \oint E \cdot dA \cos 0^\circ + \oint E \cdot dA \cos 90^\circ = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow \oint E \cdot dA = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow E \oint dA = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow EA = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow E \cdot 2\pi r l = \frac{\int_V \rho \cdot dV}{\epsilon_0}$$

$$\Rightarrow E \cdot 2\pi r l = \frac{\int \rho \pi r^2 l}{\epsilon_0}$$

$$\Rightarrow E = \frac{\int \rho r}{2\epsilon_0} \quad [r = \text{radius of gaussian surface}]$$

So, the electric field inside the thick conducting rod is $\frac{\int \rho r}{2\epsilon_0}$

Ans

⑥ In the outside of the thick conducting rod, $r > R$.

We know that,

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow \oint E \cdot dA \cos 90^\circ + \oint E \cdot dA \cos 0^\circ + \oint E \cdot dA \cos 90^\circ = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow \oint E \cdot dA = \frac{q_{enc}}{\epsilon_0}$$

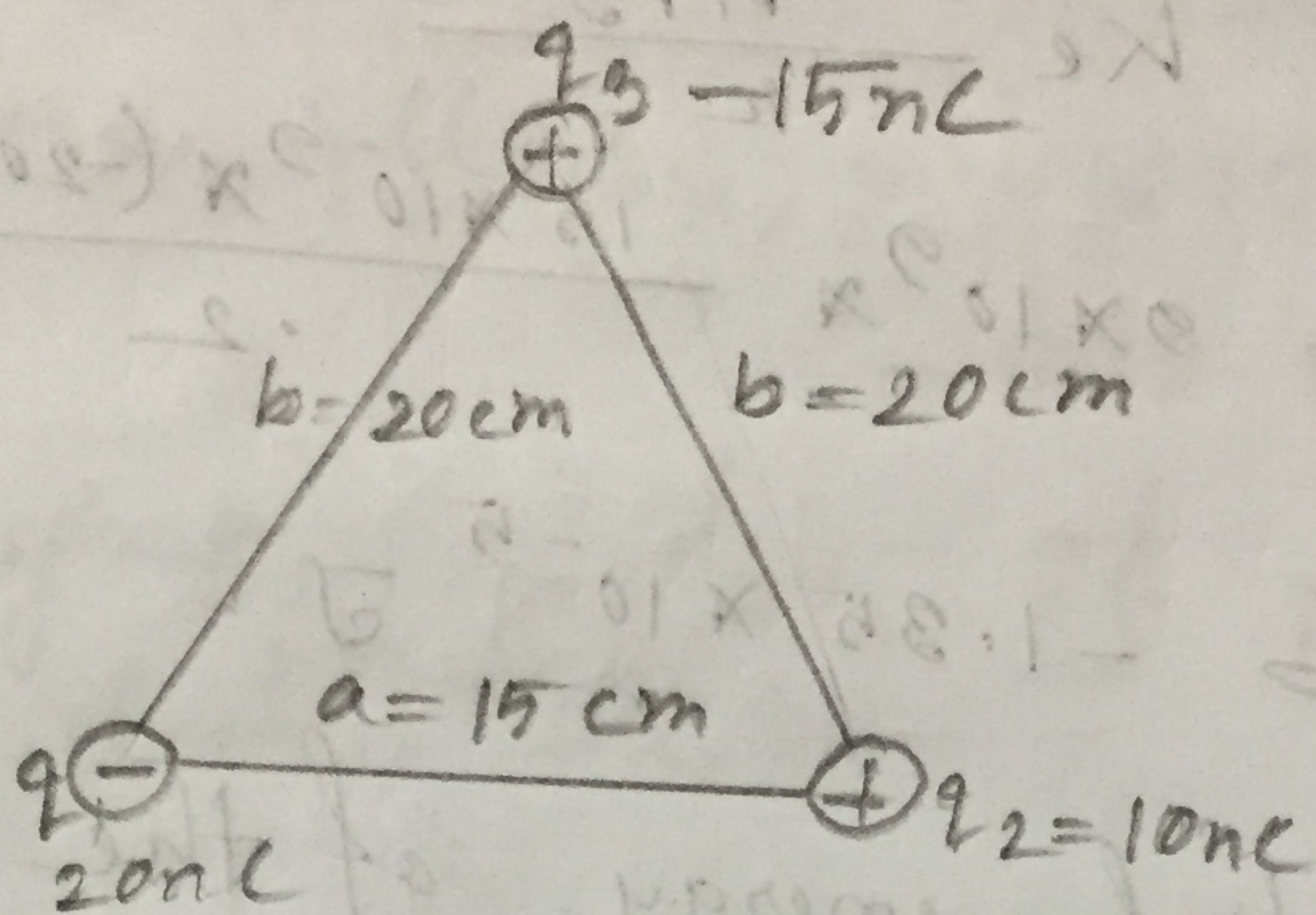
$$\Rightarrow EA = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow E \cdot 2\pi r l = \frac{\int r R^2 L}{\epsilon_0}$$

$$\Rightarrow E = \frac{\int R^2}{2 \epsilon_0 r}$$

Ahe

Ans. to Q. No. 2



Q. Ans:

Given that,

$$q_1 = -20 \text{ nC} = -20 \times 10^{-9} \text{ C}$$

$$q_2 = 10 \text{ nC} = 10 \times 10^{-9} \text{ C}$$

$$q_3 = -15 \text{ nC} = -15 \times 10^{-9} \text{ C}$$

$$a = 15 \text{ cm} = 0.15 \text{ m}$$

$$b = 20 \text{ cm} = 0.2 \text{ m}$$

1st,

Potential energy between q_1 and q_2 ,

$$\begin{aligned} U_{12} &= k_e \frac{q_1 q_2}{r} \\ &= 9 \times 10^9 \times \frac{(-20 \times 10^{-9}) \times (10 \times 10^{-9})}{0.15} \\ &= -1.2 \times 10^{-5} \text{ J} \end{aligned}$$

q_2 and q_3 ,

$$\begin{aligned} U_{23} &= k_e \frac{q_2 q_3}{r} \\ &= 9 \times 10^9 \times \frac{10 \times 10^{-9} \times 15 \times 10^{-9}}{0.2} = 6.75 \times 10^{-6} \text{ J} \end{aligned}$$

q_1 and q_3 .

$$\begin{aligned} U_{13} &= k_e \frac{q_1 q_3}{r} \\ &= 9 \times 10^9 \times \frac{15 \times 10^{-9} \times (-20 \times 10^{-9})}{.2} \\ &= -1.35 \times 10^{-5} \text{ J.} \end{aligned}$$

$\therefore$ Potential energy of the whole system

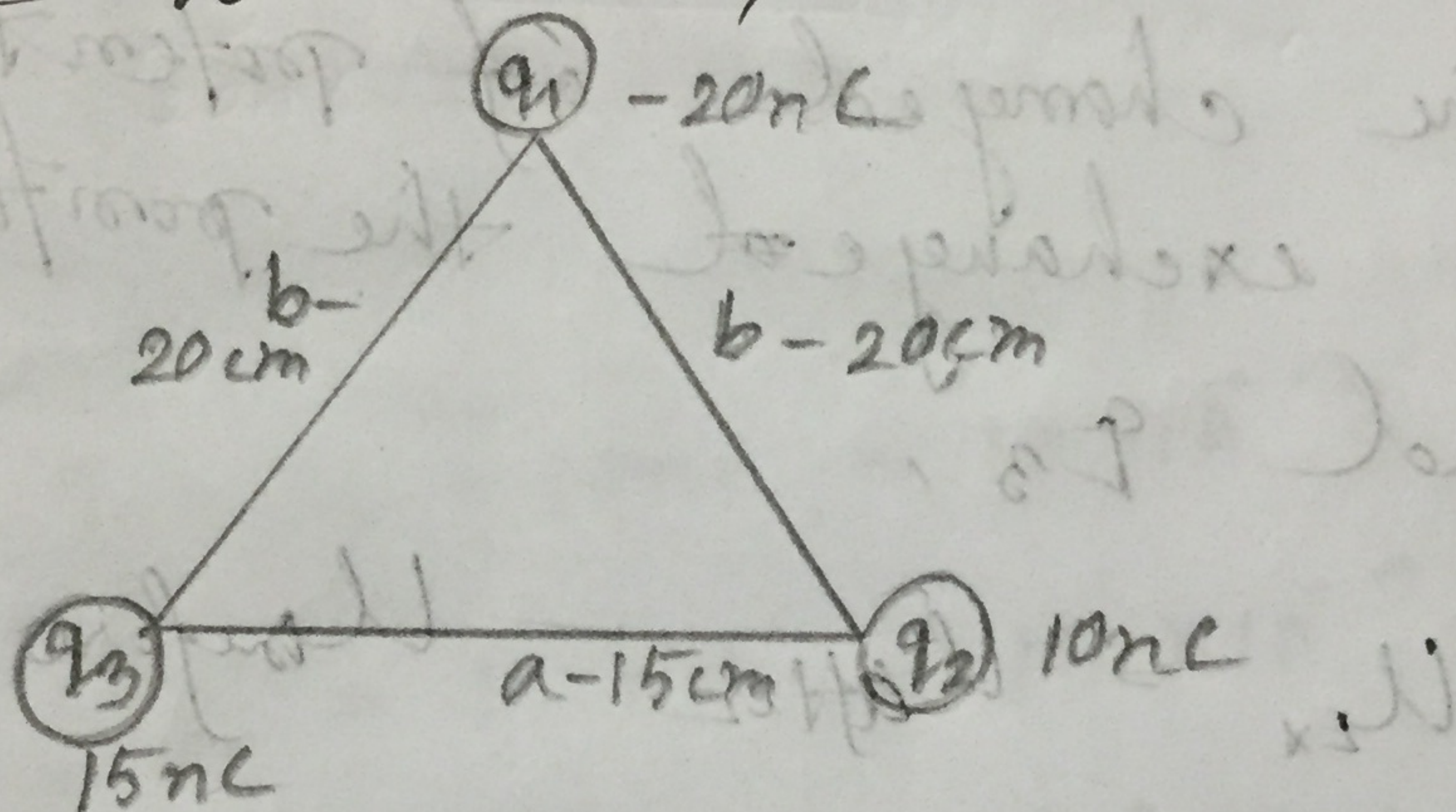
$$\begin{aligned} U &= U_{12} + U_{23} + U_{13} \\ &= (-1.2 \times 10^{-5} + 6.75 \times 10^{-6} - 1.35 \times 10^{-5}) \\ &= -1.872 \times 10^{-5} \text{ J} \end{aligned}$$

Ans

(b) Ans:

from the answer a, Potential energy before, $U_{\text{before}} = -1.872 \times 10^{-5} \text{ J}$
 $= -18.72 \mu\text{J}.$

After changing the position of q_1 and q_3 , the figure looks like,



$$U_{12} = k_e \frac{q_1 q_2}{r} \\ = 9 \times 10^9 \times \frac{(-20 \times 10^{-9}) \times (10 \times 10^{-9})}{.2} \\ = -9 \times 10^{-6} \text{ J}$$

$$U_{23} = k_e \frac{q_2 q_3}{r} \\ = 9 \times 10^9 \times \frac{(10 \times 10^{-9}) \times (15 \times 10^{-9})}{.15} \\ = 9 \times 10^{-6} \text{ J}$$

$$U_{13} = k_e \frac{q_1 q_3}{r} \\ = 9 \times 10^9 \times \frac{(-20 \times 10^{-9}) \times (15 \times 10^{-9})}{.2} = -1.35 \times 10^{-5} \text{ J}$$

$$\begin{aligned}
 U_{\text{after}} &= U_{12} + U_{23} + U_{13} \\
 &= \left\{ (-9 \times 10^{-6}) + (9 \times 10^{-6}) + (-1.35 \times 10^{-5}) \right\} \text{ J} \\
 &= -1.35 \times 10^{-5} \text{ J} \\
 &= -13.5 \mu\text{J}
 \end{aligned}$$

So, the changed of potential energy after exchanged the position of q_1 and q_3 ,

$$U_{\text{ex}} = U_{\text{after}} - U_{\text{before}}$$

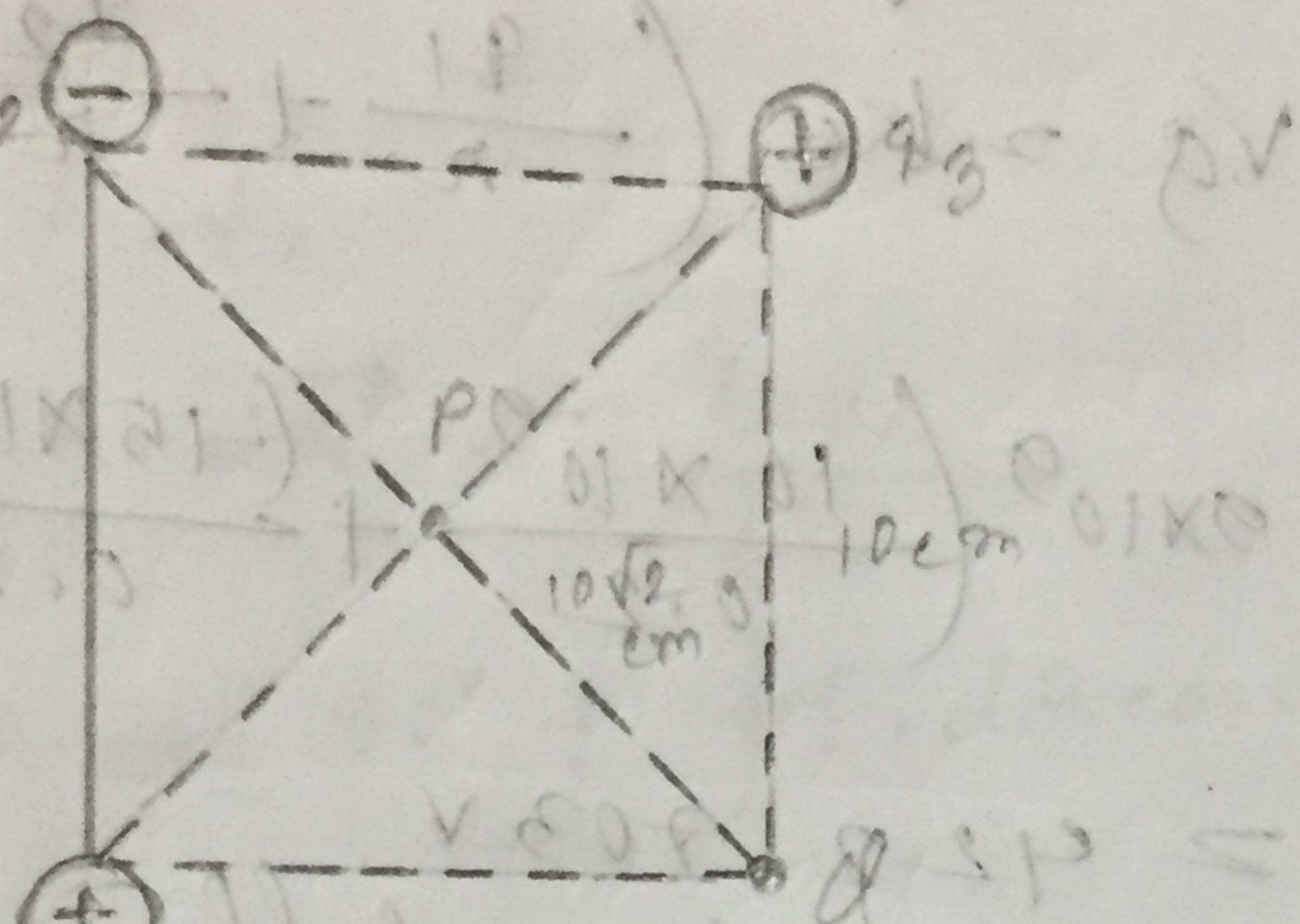
$$= (-13.5 \mu\text{J}) - (-18.7 \mu\text{J})$$

$$= \cancel{5.25 \times 10^{-6}}$$

$$= 5.25 \mu\text{J}$$

Ans

Ans. to Q. 03



⇒ given that,

$$q_1 = q_3 = 10 \text{ nC} = 10 \times 10^{-9} \text{ C}$$

$$q_2 = 15 \text{ nC} = 15 \times 10^{-9} \text{ C}$$

Length of each side of square = 20 cm = 0.2 m.

$$r = 10\sqrt{2} \text{ cm} \\ = 0.141 \text{ m}$$

Potential energy at the point P,

$$V_p = k_e \left(\frac{q_1}{r} + \frac{q_2}{r} + \frac{q_3}{r} \right) \\ = 9 \times 10^9 \left[\frac{10 \times 10^{-9}}{0.141} + \frac{(-15 \times 10^{-9})}{0.141} + \frac{10 \times 10^{-9}}{0.141} \right]$$

$$= 318.108 \text{ V}$$

Potential energy at the point G,

$$V_G = k_e \left(\frac{q_1}{r} + \frac{q_2}{r} + \frac{q_3}{r} \right)$$

$$= 9 \times 10^9 \left(\frac{10 \times 10^{-9}}{0.2} + \frac{(-15 \times 10^{-9})}{0.2} + \frac{10 \times 10^{-9}}{0.2} \right) \text{ V}$$

$$= 422.703 \text{ V}$$

So, the potential difference between point P and G is, $= V_P - V_G$

$$= (318.198 - 422.703) \text{ V}$$

$$= -104.505 \text{ V}$$

A